ICE Task 3: Final Report

What customers are telling us

Themes and sentiment in Amazon food product reviews

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At a glance

393,888	95.4%	82%	20.7%
reviews analyzed after cleaning (from 568,454)	of reviews labelled positive or negative correctly	of unhappy reviews found by the model	negative rate when packaging is mentioned, against 14.5% overall

Overall mood is good, but it is slipping. 78% of reviews are positive (4 or 5 stars). The share of one- and two-star reviews rose from 8.3% in 2005 to 16.1% in 2012.

Packaging, freshness, price and delivery are where customers get unhappy. Reviews that mention packaging or damage are the most likely to be negative. Taste is the biggest pool of complaints by volume, even though a taste mention on its own is less risky than average.

The sentiment model is accurate enough to monitor reviews at scale. It reads sentiment correctly 95.4% of the time on reviews it has never seen. Its weak spot is mixed reviews, where a customer loves the food and hates the delivery. Details and recommendations follow.

1. Introduction

The client sells food products on Amazon and has seen customer feedback become more varied. Management wants to know what customers are happy or unhappy about, and how customers feel about the products overall.

I answered both questions from the review text itself. Two tools do the work. A **topic model** reads the reviews without any hints and groups them by what they talk about. A **sentiment model** learns from the star ratings which words signal a happy or an unhappy customer and then reads new reviews at scale. I also counted how often reviews mention specific business topics (taste, packaging, delivery, price, freshness) and how risky each one is.

The report follows the analysis in order: the data, the clean-up, how text became numbers, how the models were trained and checked what I found, and what I recommend. A glossary of the technical words is at the end.

2. Dataset overview

I used the public Amazon Fine Food Reviews dataset. It holds 568,454 reviews of 74,258 products by 256,059 reviewers, written between 8 October 1999 and 26 October 2012. Each review has the text, a short summary line, a star rating from 1 to 5 and a date.

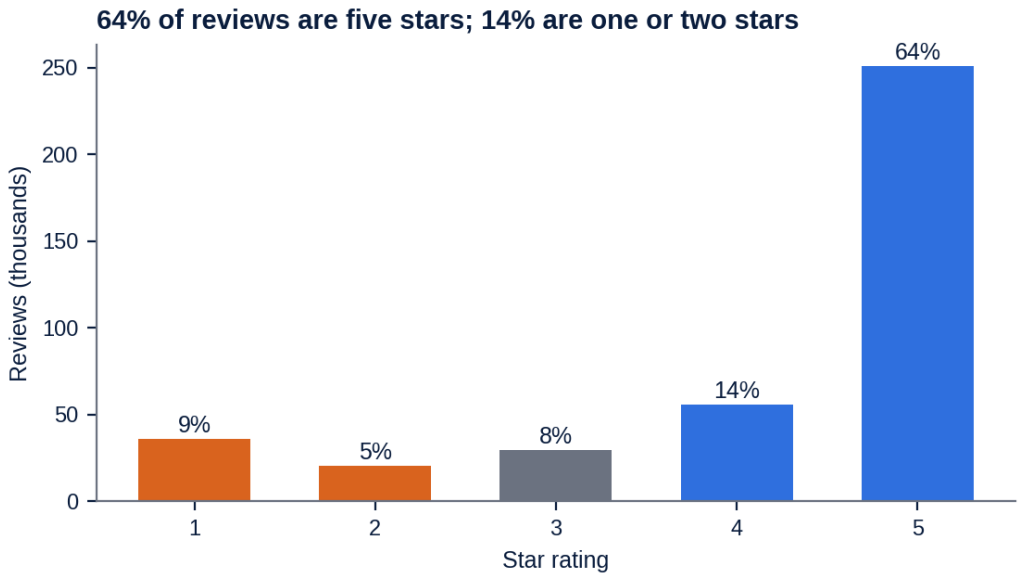


Figure 1: Star ratings after cleaning. Most reviews are five stars, which is typical of online reviews.

For modelling, I treat 4 and 5 stars as **positive** (307,017 reviews) and 1 and 2 stars as **negative** (57,103 reviews). Three-star reviews (29,768) are mixed, so I kept them out of training and scored them afterwards. Negative reviews make up 15.7% of the reviews with a clear sentiment. That imbalance is why accuracy alone cannot be trusted later.

> **What this data can and cannot tell us**
>
> It describes **people who chose to write a review**, and they lean towards very happy or very unhappy. It does not describe every customer.
>
> The data stops in 2012. Products, packaging and customer habits may have changed since.
>
> There are product IDs but no product names, so I can say what customers complain about, but not which brand or courier is to blame.
>
> Reviewer names and IDs are personal data. I did not use names and I report only totals.

3. Data cleaning and exploration

3.1 What I cleaned

Step	Rows affected	What it means
Starting point	568,454	Rows in the file
Duplicate reviews removed	174,562	Same reviewer, same time, same text: one review copied across product variants (30.7% of the file)
Impossible helpfulness counts removed	2	More helpful votes than total votes
Summaries missing in the raw file	27	Review text was present, so I kept those reviews with an empty summary
Empty after text clean-up	2	Nothing left once filler words were removed
Reviews analyzed	**393,888**	Used for everything that follows

Text clean-up. Every review was turned to lowercase, stripped of web links and HTML code, and stripped of punctuation and digits. Short forms like "didn't" became "did not". Common filler words

("the", “and”, "is") were removed. I kept words like "not", "no" and "never" on purpose. "Not good" and "good" mean opposite things, and a model that loses the "not" will get them backwards.

3.2 What the reviews look like

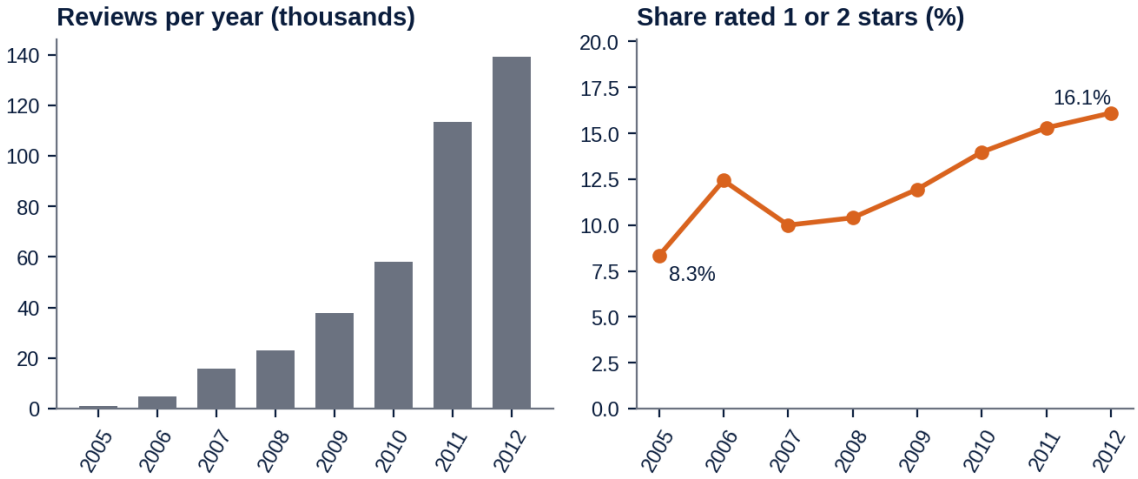


Figure 2: Review volume has grown sharply, and the share of one- and two-star reviews has risen with it.

Satisfaction is drifting down. The share of one- and two-star reviews moved from 8.3% in 2005 to 16.1% in 2012, while the average rating went from 4.50 to 4.11. Volume grew from 1,069 reviews a year to 139,178, so some of this may simply be a wider customer base.

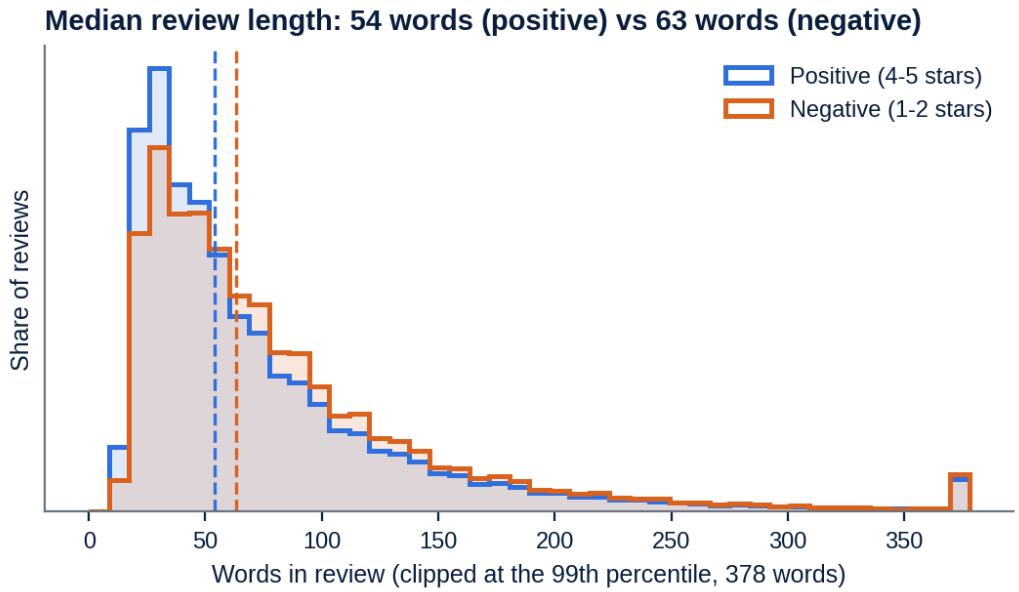


Figure 3: Review length in words. The last bar holds every review longer than the 99th percentile. Dashed lines mark each group's median.

Unhappy customers write more. The median negative review is 63 words, and the median positive review is 54. Across all reviews the median is 57 words and the mean is 80, pulled up by a few very long ones.

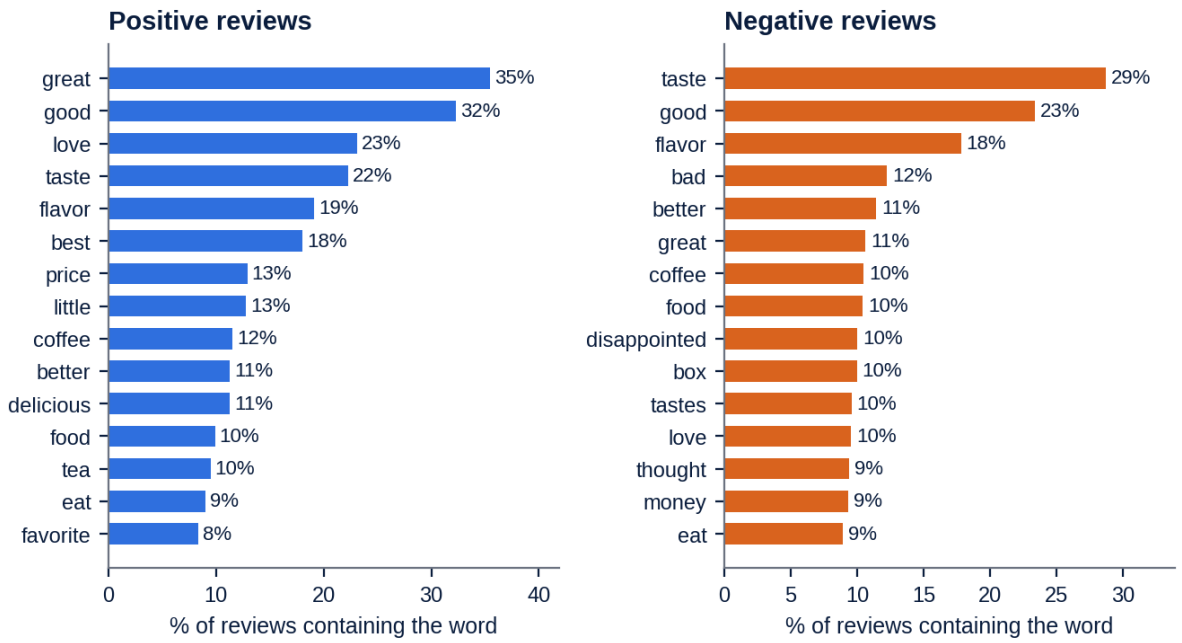


Figure 4: The most common words in positive and negative reviews. Negations and everyday filler words are hidden in this chart only. They stay in the model.



Figure 5: Word clouds. A bigger word appears in more reviews.

Both groups talk about the same things: taste, flavor, coffee, food. Common words alone do not separate happy from unhappy customers. Even "good" appears in 23% of negative reviews, so a single word cannot tell a happy review from an unhappy one. The next chart looks at the words that are far more common in one group than in the other.

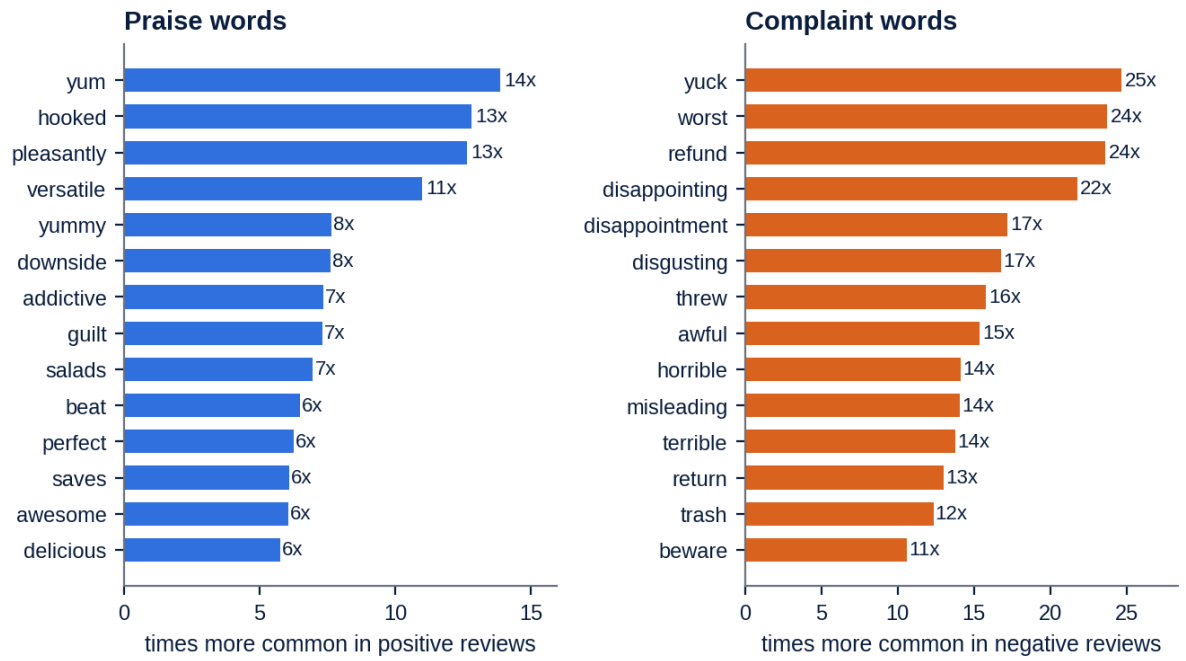


Figure 6: Words that set praise apart from complaints (only words that appear in at least 1,000 reviews). "14x" means 14 times more common in that group.

The complaint words are blunt and strongly negative (yuck, worst, refund, disappointing, disappointment). The praise words are warm and enthusiastic (yum, hooked, pleasantly, versatile, yummy).

4. Feature engineering

A computer cannot read sentences, so every review had to become a row of numbers first. I tried three ways of doing that:

- **Bag of words:** count how many times each word appears.
- **TF-IDF:** count words, then turn down the volume on words that appear in almost every review (like "product") so rarer, more telling words stand out.
- **Phrases (n-grams):** also count pairs of neighbouring words, so "not good" and "too sweet" become things the model can learn.

I compared four versions on a slice of the training reviews, using the same simple classifier for each, so the test reviews stayed untouched.

Way of turning text into numbers	Columns	Overall score (macro-F1)	Score on unhappy reviews (F1)
Bag of words, single words (20k)	18,160	0.862	0.772
TF-IDF, single words (20k)	18,160	0.866	0.780
TF-IDF, 1-2-word phrases (20k)	20,000	0.893	0.822
TF-IDF, 1-2-word phrases (100k)	100,000	0.903	0.837

What I chose and why. TF-IDF beat plain counts, and adding two-word phrases gave the biggest jump (overall score 0.866 to 0.893 at the same size). A bigger vocabulary helped a little more (0.903). So, the improved model uses TF-IDF with one- and two-word phrases and a vocabulary cap of 200,000 (the full training set holds more phrases than the slice I compared on). The phrases matter most for negation: the model learns that "not disappointed" is praise and "not worth" is a complaint.

5. Model training

I split the 364,120 reviews with a clear sentiment into **291,296 for training** (80%) and **72,824 for testing** (20%). Both sets keep the same mix of happy and unhappy reviews (15.7% and 15.7% negative). The test reviews were never used to choose any setting. All choices were made on a slice of the training data.

Model	Text to numbers	Settings
Majority-class guess	None	Always says "positive". Shows what accuracy alone hides.
Naive Bayes (baseline)	TF-IDF, single words, 20,000 columns	Smoothing 1.0 (default)
Logistic Regression (baseline)	TF-IDF, single words, 20,000 columns	Strength of penalty C = 1.0. Settings fixed before running.
Naive Bayes (tuned)	TF-IDF, 1-2 word phrases, 200,000 columns	alpha=0.05
Linear SVM (tuned)	TF-IDF, 1-2 word phrases, 200,000 columns	C=1, class_weight=balanced
Logistic Regression (improved)	TF-IDF, 1-2 word phrases, 200,000 columns	**C=20, class_weight=None**

For tuning I tried several values of each setting on a validation slice and kept the best one by overall score. Logistic Regression is the model I carry forward because it gives a probability for each review and its word weights can be read and explained. A neural network was left out because training one on this many reviews needs more computing power than I had, and I have not tested one.

6. Model evaluation

6.1 Starting point: the baseline

The baseline Logistic Regression scored 94.2% accuracy, which sounds excellent. It is less impressive than it looks. A model that always answers "positive" scores 84.3% here, because most reviews are positive. What matters to the business is finding the unhappy customers, and on that the baseline found 73.8% of them. Roughly one in four complaints slipped through.

6.2 Improvement: what changed, why and what it did

What changed	Why	Effect
Single words became 1-2 word phrases	Negation and short phrases ("not good", "too sweet") carry the meaning that single words lose	Biggest single gain in the feature comparison (overall score 0.866 to 0.893)
Vocabulary grew from 20,000 to 200,000 columns	Many telling phrases are rare, and a small vocabulary throws them away	Further gain in the comparison (0.893 to 0.903 at 100,000)
Penalty loosened (C from 1 to 20)	With a lot of data and many features, a gentler penalty lets the model use more of what it sees	Chosen by the validation slice, not by the test set
Class weighting was tested	Only about 1 in 6 labelled reviews is negative, so the model could lean towards "positive"	The tuning did not choose it for the final model. It helped at low penalty values but not at the final one

Result on the held-out test reviews	Baseline	Improved	Change
Accuracy	94.2%	95.4%	+1.2 points
Overall score (macro-F1)	0.883	0.910	+0.028
Unhappy reviews found (recall)	73.8%	82.1%	+8.3 points
Flags that were right (precision, negative)	87.2%	87.7%	+0.5 points
Score on unhappy reviews (F1)	0.799	0.848	+0.049
Ranking ability (ROC AUC)	0.973	0.981	+0.008

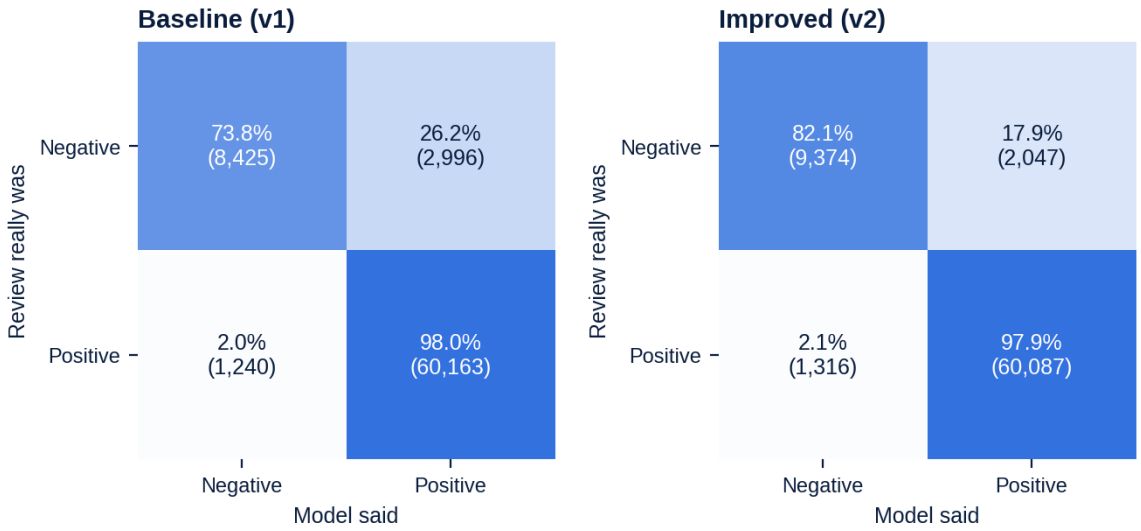


Figure 7: What the model said against what the review really was. Percentages are of each real group.

The improved model beats the baseline on every measure, which meets the test I set in advance. It now finds 82.1% of unhappy reviews instead of 73.8%, and it did that without flagging more happy customers by mistake (precision 87.7% against 87.2%).

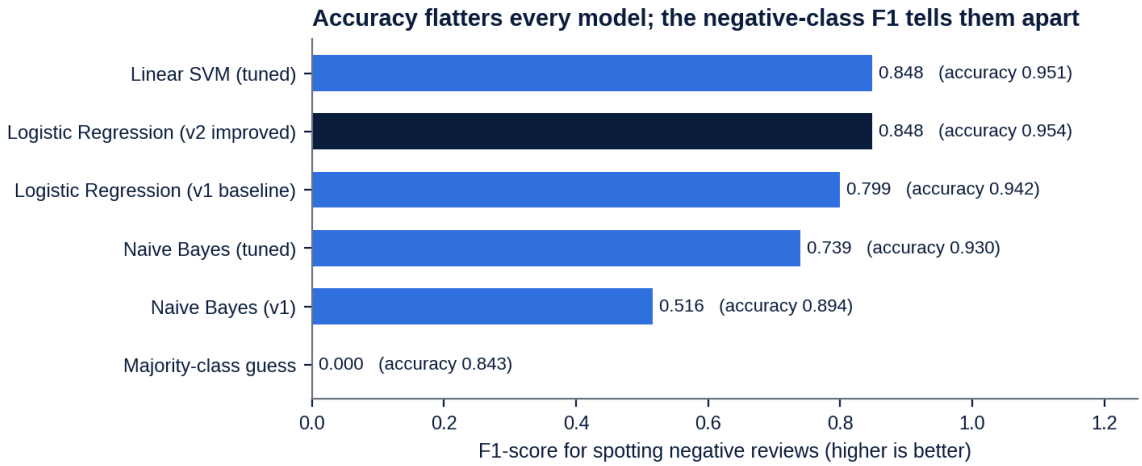


Figure 8: Every model compared on the metric the business cares about most.

A close call. The tuned Linear SVM ties the improved Logistic Regression on the score for unhappy reviews (0.848 against 0.848). It finds more complaints (87.8%) but a smaller share of its flags are right (82.1%). I stayed with Logistic Regression because it gives a probability for each review and its reasoning can be shown to the client. If missing a complaint costs more than reading a few extra false alarms, the SVM is a fair alternative.

6.3 Strengths, weaknesses and what it means for the business

Strengths	Weaknesses	Business meaning
- 95.4% accuracy and 0.981 AUC on reviews it never saw - Handles negation: "not disappointed" counts as praise - Light on computing: the final model trained in under a minute on a two-core machine once the text was converted - Shows its reasoning through word weights	- Misses about 18% of unhappy reviews - Two-star reviews are hardest: 27% wrong, against 12% for one star and 1% for five stars - Reads words, not meaning, so sarcasm and mixed reviews trip it up	- Good for tracking trends and spotting spikes in complaints across thousands of reviews - Not safe as the only judge of one customer's review - Send borderline and two-star reviews to a person

What the mistakes look like. I read a random sample of them. The most common mistake is a mixed review. One customer said the beans were excellent and would be five stars, then gave one star because every can arrived dented. The model sees "excellent" and calls it praise. Others praised the food and complained about price, shipping or expiry dates. The reverse also happens: one customer says the chips are so tart they cannot eat them alone, yet still gives four stars.

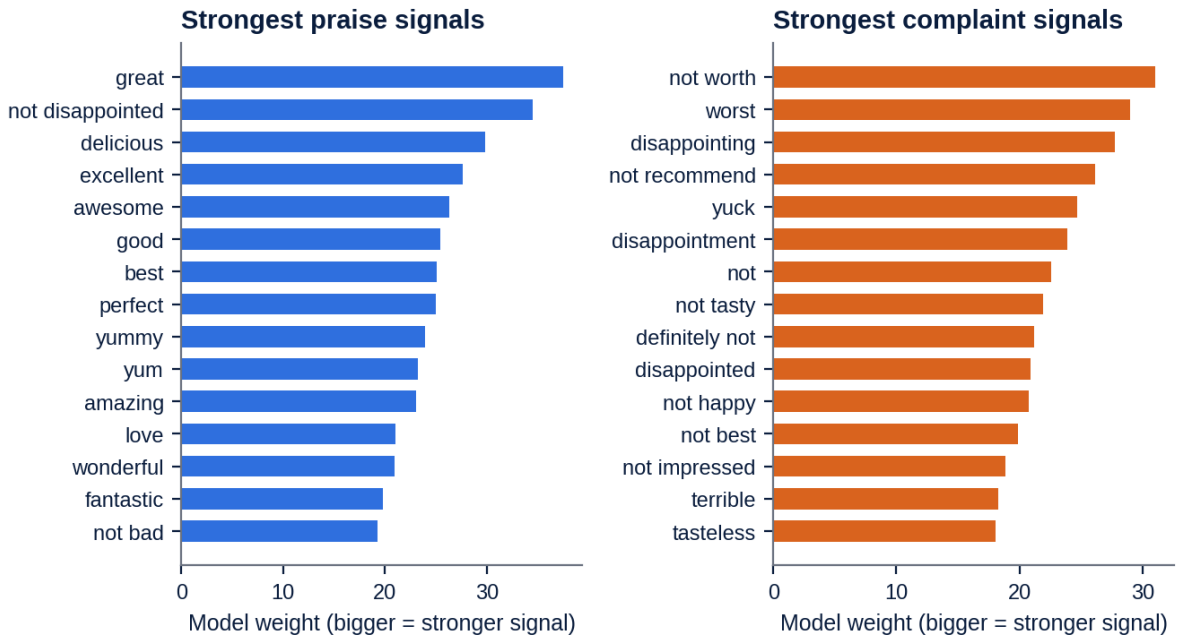


Figure 9: The words that push the improved model most strongly towards positive (blue) or negative (orange).

Two extra checks

Does it hold up on newer reviews? I trained on the oldest 80% of reviews and tested on the newest 20% (reviews from 15 May 2012 onwards). Accuracy was 95.3% and the score on unhappy reviews was 0.861, in line with the main test. I see no sign of the model going stale, though the newest window is only a few months long.

Where do three-star reviews land? The model never saw them in training. It calls 60% of the 29,768 three-star reviews positive, so three stars lean positive in the wording customers use.

7. Key findings

7.1 Overall satisfaction

78% of reviews are positive, 8% are neutral and 14% are negative. Satisfaction is healthy overall but the share of negative reviews has been climbing since 2005 (8.3% to 16.1% in 2012).

7.2 Common complaints: which topics are risky

I counted how many reviews mention each business topic and how many of those are negative. Across all reviews, 14.5% are one or two stars. A topic above that line is a problem area.

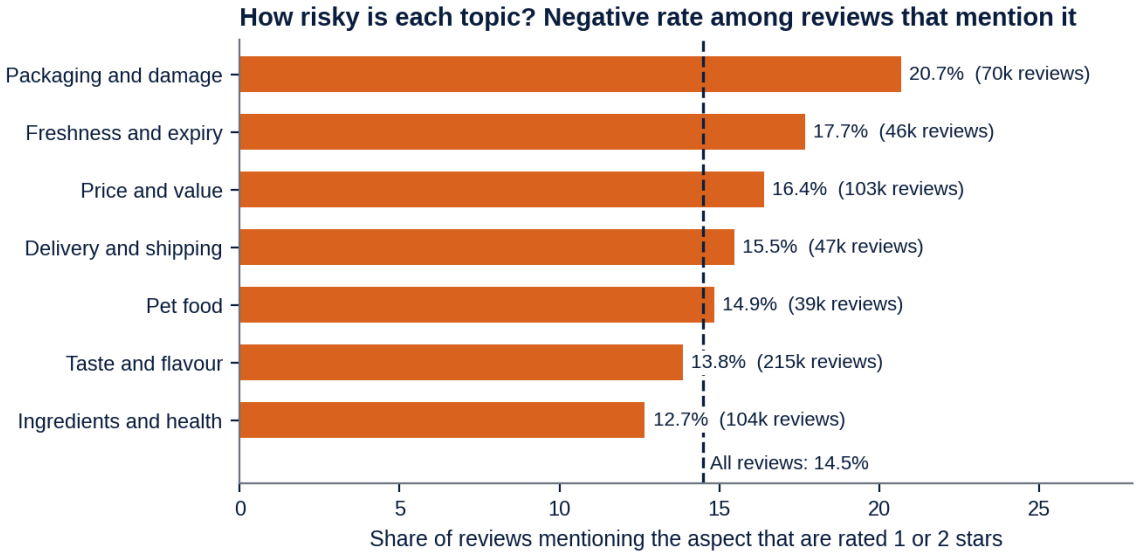


Figure 10: Share of reviews that are one or two stars among those that mention each topic. Topics are found by keyword lists (see the notebook).

Topic	Reviews mentioning it	Negative rate	Negative reviews	Positive rate
Packaging and damage	69,695	20.7%	14,418	69.3%
Freshness and expiry	46,363	17.7%	8,202	75.0%
Price and value	103,231	16.4%	16,916	75.6%
Delivery and shipping	47,352	15.5%	7,328	77.6%
Pet food	38,825	14.9%	5,766	78.1%
Taste and flavour	214,638	13.8%	29,724	77.7%
Ingredients and health	104,151	12.7%	13,196	79.6%

Packaging and damage is the highest-risk topic. 20.7% of reviews that mention it are negative, against 14.5% overall. Freshness and expiry (17.7%), price and value (16.4%) and delivery and shipping (15.5%) are also above the line.

Taste is the biggest pool of complaints by volume. A taste mention is less risky than average (13.8%), but so many reviews talk about taste that it still accounts for about 29,724 negative reviews, more than any other topic.

A caution: these counts show that reviews mention a topic, not that the topic caused the bad rating. A customer may write "great taste, but the box was crushed" and the review counts under both.

7.3 Themes the topic model found

The topic model grouped negative and positive reviews separately. Each group came out with ten topics, chosen by a score that checks whether a topic's top words really appear together. I named each topic from its leading words, so the names are my reading and not something the model produced.

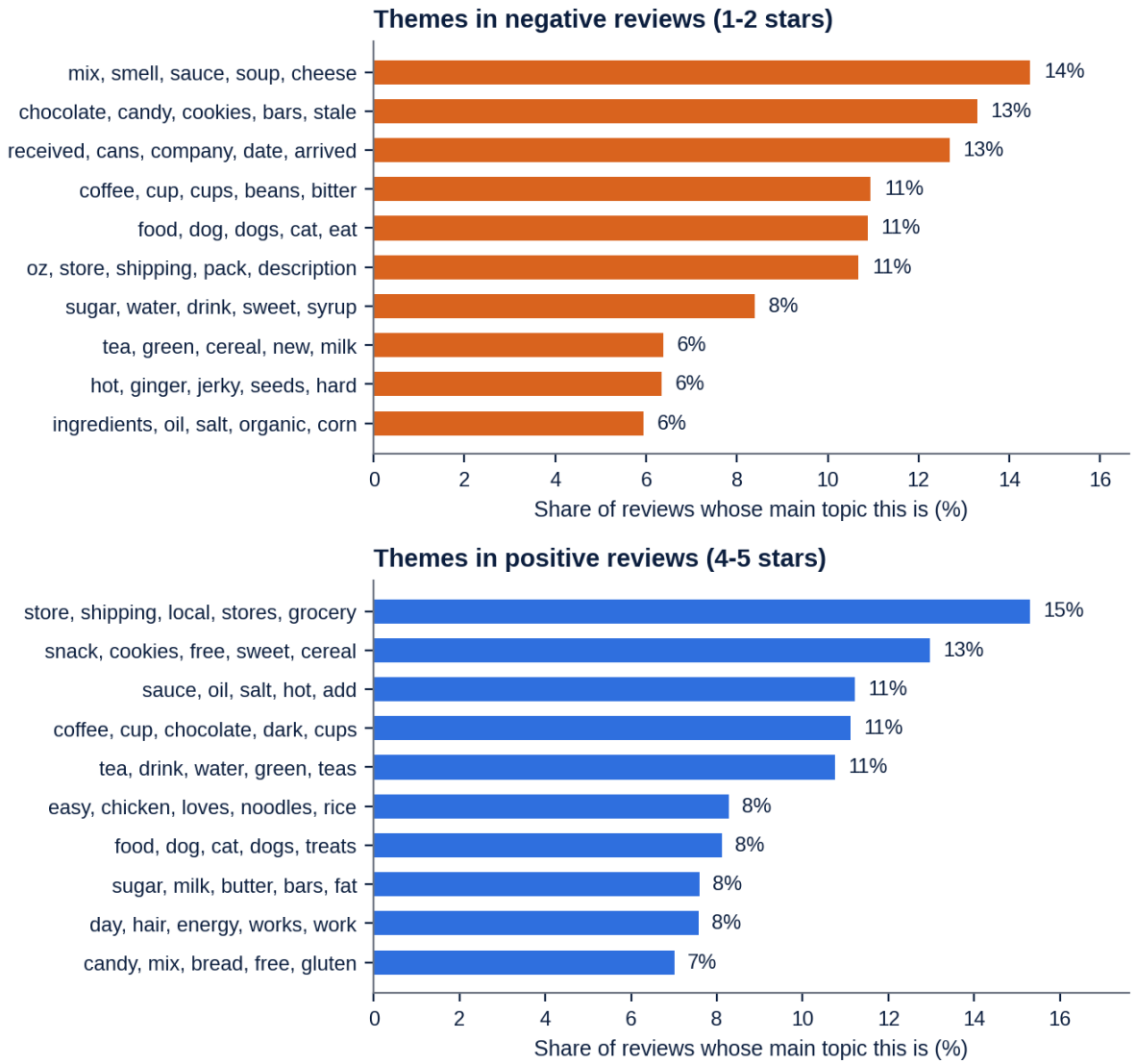


Figure 11: Topics in negative and positive reviews, each shown by its five leading words, with the share of reviews whose main topic it is.

Concern themes (negative reviews)

Theme	Share	Leading words
Mixes, sauces and gluten-free foods: smell and taste	14%	mix, smell, sauce, soup, cheese, tasted, free, stuff
Stale or hard sweets and snacks	13%	chocolate, candy, cookies, bars, stale, bar, hard, chips
Delivery, packaging and customer service	13%	received, cans, company, date, arrived, opened, packaging, customer
Coffee: bitter or weak	11%	coffee, cup, cups, beans, bitter, vanilla, weak, roast
Pet food (leading words include china and chicken)	11%	food, dog, dogs, cat, eat, treats, cats, china
Listing mismatch: size, count and description	11%	oz, store, shipping, pack, description, picture, size, received

Theme	Share	Leading words
Drinks: artificial sweetness and aftertaste	8%	sugar, water, drink, sweet, syrup, juice, artificial, bottle
Tea and cereal (mentions of new or changed versions)	6%	tea, green, cereal, new, milk, teas, changed, black
Jerky, ginger and spicy foods: too hot or too hard	6%	hot, ginger, jerky, seeds, hard, pepper, beef, licorice
Ingredients and health claims	6%	ingredients, oil, salt, organic, corn, natural, hair, fat

Praise themes (positive reviews)

Theme	Share	Leading words
Convenience: delivery and buying online instead of the local shop	15%	store, shipping, local, stores, grocery, arrived, gift, happy
Snacks: healthy and gluten-free	13%	snack, cookies, free, sweet, cereal, eat, delicious, chips
Sauces, oils and cooking ingredients	11%	sauce, oil, salt, hot, add, soup, popcorn, coconut
Coffee: smooth and strong	11%	coffee, cup, chocolate, dark, cups, strong, vanilla, smooth
Tea and drinks	11%	tea, drink, water, green, teas, sweet, sugar, drinking
Easy meals, noodles and baby food	8%	easy, chicken, loves, noodles, rice, cheese, eat, chew
Pet food that dogs and cats love	8%	food, dog, cat, dogs, treats, loves, cats, eat
Low-calorie and protein bars	8%	sugar, milk, butter, bars, fat, low, bar, chocolate
Energy and wellness products (mixed topic)	8%	day, hair, energy, works, work, using, bottle, feel
Baking mixes, candy and honey	7%	candy, mix, bread, free, gluten, delicious, beans, cake

Reading the themes. Complaints are not only about taste. About 37% of the negative reviews sit in three themes about the product arriving in good condition and as described: delivery and packaging, listings that do not match what arrived, and stale or hard goods. That fits the keyword counts above. On the praise side, customers value convenience (buying online rather than at a local shop), snacks that are healthy or gluten-free, and a smooth, strong cup of coffee.

8. Recommendations

In priority order. Each one points back to the evidence above.

Priority	Action	Evidence
1 (high)	**Fix packaging first.** Review packing for cans, bottles and other heavy or fragile items, ask customers to report damage, and track the share of reviews that mention damage each month.	Highest negative rate of any topic (20.7% against 14.5%). Dented cans came up in the model's own mistakes.
2 (high)	**Tighten freshness control.** Set a minimum shelf life at dispatch, rotate stock oldest first, and show the best-before date on the listing.	Freshness and expiry runs at 17.7% negative. Stale or hard sweets and snacks are one of the largest concern themes.
3 (high)	**Make listings match the product.** Check size, count and flavour in titles and photos, and show the price per unit.	A whole concern theme is about size, count, description and picture. Price and value is 16.4% negative across 103,231 reviews.

Priority	Action	Evidence
4 (medium)	**Use reviews as product quality control.** Send repeat taste complaints to suppliers: bitter or weak coffee, artificial aftertaste in drinks.	Taste accounts for about 29,724 negative reviews, the largest pool.
5 (medium)	**Watch delivery and service.** Look at the carriers and fulfilment routes behind delivery complaints. This dataset has no courier data, so that link needs the client's own records.	Delivery and shipping is 15.5% negative, and a concern theme covers delivery, packaging and customer service.
6 (medium)	**Run the sentiment model as a monitor.** Score new reviews every week, track the negative share by product category, and alert on spikes. Send two-star and borderline reviews to a person.	95.4% accuracy, finds 82% of unhappy reviews, and is weakest on two-star reviews (27% wrong).
7 (low)	**Play to the praise.** Lead listings with what customers love: convenience, healthy and gluten-free snacks, smooth strong coffee.	These are the leading praise themes and the praise words in Figure 6.
8 (low)	**Check where pet food is made.** The word china appears among the leading words of the pet food complaints. Confirm whether origin is a real concern and state it clearly on listings.	A pattern in the topic words only. Needs checking against actual reviews before acting.

Next step if more accuracy is needed: a Transformer-based model reads word order and context, so it should handle sarcasm and mixed reviews better. It needs more computing power and I have not tested it.

Appendix A: Settings and files

Item	Value
Train/test split	80/20, stratified, random seed 42. Train 291,296, test 72,824
Text features (baseline)	TF-IDF, single words, min_df 5, max_features 20,000, log-scaled term counts
Text features (improved)	TF-IDF, 1 to 2 word phrases, min_df 5, max_features 200,000, log-scaled term counts
Tuning	Slice of about 120,000 training reviews, 80/20 inside it. Score used: macro-F1
Best settings found	Linear SVM: C=1, class_weight=balanced. Logistic Regression: C=20, class_weight=None. Naive Bayes: alpha=0.05
Topic model	LDA, online learning, 8 passes, vocabulary 6,000 words, 40,000 reviews sampled per group. Topic counts tried 4, 6, 8, 10; chosen by NPMI coherence
Code	ICE3_Analysis.ipynb (run top to bottom). The same code is in ICE3_pipeline.py. Results are saved in results.json

Appendix B: Limits of this study

- Reviews come from people who chose to write one. They are not a random sample of customers.
- Star ratings stand in for sentiment. Some reviews do not match their stars.
- The data ends in 2012. Check findings against current reviews before acting.
- Topic names are my reading of each topic's leading words. Keyword topic counts show mentions, not causes.
- 582 rows still share their exact text with another row (from different reviewers or dates). I kept them.

- The model has no sense of sarcasm and can misread mixed reviews. Use it for trends, not for judging one customer.
- The data has no information on reviewers' backgrounds, so I could not test whether the model works equally well for different groups of customers.

Appendix C: Glossary

Term	Plain meaning
Sentiment	Whether a review is positive or negative in tone.
Topic model (LDA)	A method that reads many reviews and groups them by the subjects they talk about, without being told the subjects in advance.
TF-IDF	A way of scoring words so that words common in one review but rare overall count for more.
Phrase (n-gram)	A run of neighbouring words, such as "not good", treated as one unit.
Accuracy	The share of reviews the model labelled correctly.
Precision	When the model says "negative", how often it is right.
Recall	Of all the really negative reviews, how many the model found.
F1-score	One number that balances precision and recall. 1.0 is perfect.
Macro-F1	The average of the F1 scores for positive and negative, counting both equally.
Confusion matrix	A grid showing how many reviews were labelled correctly and how many were mixed up.
ROC AUC	How well the model ranks unhappy reviews above happy ones. 0.5 is a coin flip, 1.0 is perfect.